STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-GOOD-DGA

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5759-5768 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 5770-5830, the complete author proof (59 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 8 of queue_sdga.json (pb=2440, stmt 5759-5768, proof 5770-5830).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FSH: sources/stacks-sdga/pinned-original-source.tex, source lines 3186–3196. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FU1: sources/stacks-sdga/pinned-original-source.tex, source lines 5632–5643. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSF: sources/stacks-sdga/pinned-original-source.tex, source lines 3098–3126. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{B}, d)$ be a differential graded $\mathcal{O}$-algebra. There exists a quasi-isomorphism of differential graded $\mathcal{O}$-algebras $(\mathcal{A}, d) \to (\mathcal{B}, d)$ such that $\mathcal{A}$ is graded flat and K-flat as a complex of $\mathcal{O}$-modules and such that the same is true after pullback by any morphism of ringed topoi.*

Proof. The proof is exactly the same as the first proof of Lemma 0FSH but now working with free graded algebras instead of free graded modules.

We will construct $\mathcal{A} = \operatorname{colim} \mathcal{A}_i$ as in Lemma 0FU1 by constructing

$$\mathcal{A}_0 \to \mathcal{A}_1 \to \mathcal{A}_2 \to \ldots \to \mathcal{B}$$

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Let $\mathcal{S}_0$ be the sheaf of graded sets (Remark 0FSF) whose degree n part is $\mathrm{Ker}(\mathrm{d}^n_{\mathcal{B}})$. Consider the homomorphism of differential graded modules

$$\mathcal{A}_0 = \mathcal{O}\langle \mathcal{S}_0 \rangle \longrightarrow \mathcal{B}$$

where map sends a local section s of $\mathcal{S}_0$ to the corresponding local section of $\mathcal{A}^{\deg(s)}$ (which is in the kernel of the differential, so our map is a map of differential graded algebras indeed). By construction the induced maps on cohomology sheaves $H^n(\mathcal{A}_0) \to H^n(\mathcal{B})$ are surjective and hence the same will remain true for all i.

Induction step of the construction. Given $\mathcal{A}_i \to \mathcal{B}$ denote $\mathcal{S}_{i+1}$ the sheaf of graded sets whose degree n part is

$$\mathrm{Ker}(\mathrm{d}^{n+1}_{\mathcal{A}_i}) \times_{\mathcal{B}^{n+1}, \mathrm{d}} \mathcal{B}^n$$

This comes equipped with a canonical map

$$\delta_{i+1} : \mathcal{S}_{i+1} \longrightarrow \mathcal{A}_i$$

whose image is contained in the kernel of $\mathrm{d}_{\mathcal{A}_i}$ by construction. Hence $\mathcal{A}_{i+1} = \mathcal{O}\langle \mathcal{S}_0 \amalg \ldots \mathcal{S}_{i+1} \rangle$ has a differential exteding the differential on $\mathcal{A}_i$, see discussion at the start of this section. The map from $\mathcal{A}_{i+1}$ to $\mathcal{B}$ is the unique map of graded algebras which restricts to the given map on $\mathcal{A}_i$ and sends a local section $s = (a, b)$ of $\mathcal{S}_{i+1}$ to b in $\mathcal{B}$. This is compatible with differentials exactly because $\mathrm{d}(b)$ is the image of a in $\mathcal{B}$.

The map $\mathcal{A} \to \mathcal{B}$ is a quasi-isomorphism: we have $H^n(\mathcal{A}) = \mathrm{colim}\, H^n(\mathcal{A}_i)$ and for each i the map $H^n(\mathcal{A}_i) \to H^n(\mathcal{B})$ is surjective with kernel annihilated by the map $H^n(\mathcal{A}_i) \to H^n(\mathcal{A}_{i+1})$ by construction. Finally, the flatness condition for $\mathcal{A}$ where shown in Lemma 0FU1. $\square$